

ASCEND INSTITUTE, GANGTOK

CHAPTER-WISE TEST

ICSE CLASS 8 - MATHEMATICS

Chapter: Identities

Date: _____

Name: _____

Maximum Marks: 20

Time: 40 minutes

Instructions: Answer all questions. Section A contains 8 MCQs of 1 mark each. Section B contains 6 questions of 2 marks each. Show the necessary steps wherever required.

SECTION A - 8 MARKS

1. The expansion of $(x + 4)^2$ is:

(i) $x^2 + 8x + 16$

(k) $x^2 + 16$

(j) $x^2 + 4x + 16$

(l) $x^2 - 8x + 16$

Answer: _____

2. $(3a - 2b)^2$ is equal to:

(i) $9a^2 - 12ab + 4b^2$

(k) $9a^2 + 12ab + 4b^2$

(j) $9a^2 - 6ab + 4b^2$

(l) $6a^2 - 12ab + 4b^2$

Answer: _____

3. Complete: $(2p + q)^2 = 4p^2 + \text{_____} + q^2$.

(i) $2pq$

(k) $6pq$

(j) $4pq$

(l) $8pq$

Answer: _____

4. Which expansion is incorrect?

(i) $(x + 3)^2 = x^2 + 6x + 9$

(k) $(2x + 1)^2 = 4x^2 + 2x + 1$

(j) $(x - 5)^2 = x^2 - 10x + 25$

(l) $(a - b)(a + b) = a^2 - b^2$

Answer: _____

5. $x^2 - 49$ can be written as:

(i) $(x - 7)^2$

(k) $(x - 7)(x + 7)$

(j) $(x + 7)^2$

(l) $(x - 49)(x + 1)$

Answer: _____

6. Which expression equals $(p + q)^2 - 2pq$?

(i) $p^2 + q^2$

(k) $2pq$

(j) $p^2 - q^2$

(l) $p + q$

Answer: _____

7. If $a + b = 9$ and $ab = 14$, then $a^2 + b^2 =$

(i) 39

(k) 67

(j) 53

(l) 81

Answer: _____

8. If $x - y = 6$ and $xy = 5$, then $x^2 + y^2 =$

(i) 26

(k) 46

(j) 36

(l) 56

Answer: _____

SECTION B - 12 MARKS

Answer all questions. Continue on the back side if additional space is required.

9. Expand and simplify: $(4x + 3y)^2$. [2]

10. Expand and simplify: $(5a - 2b)^2$. [2]

11. Simplify using identities: $(2x + 3)^2 - (2x - 3)^2$. [2]

12. Factorise completely: $81p^2 - 16q^2$. [2]

13. If $a + b = 11$ and $ab = 24$, find $a^2 + b^2$. [2]

14. A square has side $(2m - 3)$ cm. Find its area in simplified form. [2]

ASCEND INSTITUTE, GANGTOK

CHAPTER-WISE TEST

ICSE CLASS 8 - MATHEMATICS

Chapter: Probability

Date: _____

Name: _____

Maximum Marks: 20

Time: 40 minutes

Instructions: Answer all questions. Section A contains 8 MCQs of 1 mark each. Section B contains 6 questions of 2 marks each. Show the necessary steps wherever required.

SECTION A - 8 MARKS

1. Which of the following can be the probability of an event?

- | | |
|----------|----------|
| (i) -0.3 | (j) 0.65 |
| (k) 1.4 | (l) 2 |

Answer: _____

2. A fair coin is tossed once. The probability of getting a tail is:

- | | |
|-------------------|-------------------|
| (i) 0 | (j) $\frac{1}{4}$ |
| (k) $\frac{1}{2}$ | (l) 1 |

Answer: _____

3. A fair die is thrown once. The probability of getting an even number is:

- | | |
|-------------------|-------------------|
| (i) $\frac{1}{6}$ | (j) $\frac{1}{3}$ |
| (k) $\frac{1}{2}$ | (l) $\frac{2}{3}$ |

Answer: _____

4. A fair die is thrown once. The probability of getting a number greater than 4 is:

- | | |
|-------------------|-------------------|
| (i) $\frac{1}{6}$ | (j) $\frac{1}{3}$ |
| (k) $\frac{1}{2}$ | (l) $\frac{2}{3}$ |

Answer: _____

5. A bag has 3 red, 2 blue and 5 green balls. The probability of drawing a blue ball is:

- | | |
|--------------------|-------------------|
| (i) $\frac{1}{10}$ | (j) $\frac{1}{5}$ |
| (k) $\frac{2}{5}$ | (l) $\frac{1}{2}$ |

Answer: _____

6. The probability of an impossible event is:

- | | |
|-------|---------------------|
| (i) 0 | (j) $\frac{1}{2}$ |
| (k) 1 | (l) Cannot be found |

Answer: _____

7. If $P(E) = \frac{3}{8}$, then $P(\text{not } E) =$

- | | |
|-------------------|-------------------|
| (i) $\frac{3}{8}$ | (j) $\frac{5}{8}$ |
| (k) $\frac{3}{5}$ | (l) $\frac{8}{3}$ |

Answer: _____

8. Bag A has 2 red and 3 blue balls. Bag B has 4 red and 6 blue balls. From which bag is a red ball more likely?

- | | |
|-------------------------|------------------------|
| (i) Bag A | (j) Bag B |
| (k) Same chance in both | (l) Cannot be compared |

Answer: _____

SECTION B - 12 MARKS

Answer all questions. Continue on the back side if additional space is required.

9. A fair die is thrown once. Find the probability of getting a number greater than 4. [2]

10. A fair coin is tossed once. Find the probability of not getting a head. [2]

11. A bag contains 4 red, 3 blue and 1 green ball. Find the probability of drawing a blue ball. [2]

12. A bag contains 3 red, 2 blue and 5 green balls. Find the probability of drawing a ball that is not green. [2]

13. If $P(E) = \frac{3}{7}$, find $P(\text{not } E)$. [2]

14. Cards numbered 1 to 8 are placed in a box. One card is drawn at random. Find the probability that the number is prime. [2]

ASCEND INSTITUTE, GANGTOK
CHAPTER-WISE TEST
ICSE CLASS 8 - PHYSICS

Chapter: Light

Date: _____

Name: _____

Maximum Marks: 20

Time: 40 minutes

Instructions: Answer all questions. Section A contains 8 MCQs of 1 mark each. Section B contains 6 questions of 2 marks each. Show the necessary steps wherever required.

SECTION A - 8 MARKS

1. Which mirror is used as a rear-view mirror in vehicles because it gives a wider field of view?

- | | |
|-------------------|--------------------|
| (i) Plane mirror | (j) Concave mirror |
| (k) Convex mirror | (l) Any mirror |

Answer: _____

2. A dentist uses a mirror to see a magnified upright image of a tooth. The mirror is:

- | | |
|------------|-----------------|
| (i) Plane | (j) Concave |
| (k) Convex | (l) Glass plate |

Answer: _____

3. An object is 30 cm in front of a plane mirror. The distance between the object and its image is:

- | | |
|-----------|-----------|
| (i) 30 cm | (j) 45 cm |
| (k) 60 cm | (l) 90 cm |

Answer: _____

4. A ray makes an angle of 30 degrees with the surface of a plane mirror. Its angle of reflection is:

- | | |
|----------------|----------------|
| (i) 30 degrees | (j) 45 degrees |
| (k) 60 degrees | (l) 90 degrees |

Answer: _____

5. A concave mirror forms a virtual, erect and enlarged image when the object is placed:

- | | |
|----------|---------------------|
| (i) At C | (j) Between C and F |
| (k) At F | (l) Between F and P |

Answer: _____

6. For a concave mirror, an object placed at C forms an image that is:

- | | |
|--------------------------|----------------------------|
| (i) Virtual and enlarged | (j) Real and same size |
| (k) Real and diminished | (l) Virtual and diminished |

Answer: _____

7. When white light passes through a prism, the splitting of white light into colours is called:

- | | |
|-----------------------|----------------|
| (i) Reflection | (j) Dispersion |
| (k) Lateral inversion | (l) Diffusion |

Answer: _____

8. The left-right reversal seen in a plane mirror is called:

- | | |
|-----------------------|-------------------|
| (i) Refraction | (j) Dispersion |
| (k) Lateral inversion | (l) Magnification |

Answer: _____

SECTION B - 12 MARKS

Answer all questions. Continue on the back side if additional space is required.

9. State the two laws of reflection. [2]

10. Write any four properties of the image formed by a plane mirror. [2]

11. For a concave mirror, state the nature and size of the image when the object is (i) at C and (ii) between F and P. [2]

12. Give one use each of a concave mirror and a convex mirror, with a reason for the choice. [2]

13. What is dispersion of light? Write the sequence of colours obtained from white light through a prism. [2]

14. A boy stands 25 cm in front of a plane mirror. How far is his image from the mirror, and what is the distance between the boy and his image? [2]

ASCEND INSTITUTE, GANGTOK
CHAPTER-WISE TEST
ICSE CLASS 8 - PHYSICS

Chapter: Heat Transfer

Date: _____

Name: _____

Maximum Marks: 20

Time: 40 minutes

Instructions: Answer all questions. Section A contains 8 MCQs of 1 mark each. Section B contains 6 questions of 2 marks each. Show the necessary steps wherever required.

SECTION A - 8 MARKS

1. Heat naturally flows from:

- (i) A colder body to a hotter body
- (k) Both directions equally

- (j) A hotter body to a colder body
- (l) Only solids to liquids

Answer: _____

2. Evaporation mainly takes place:

- (i) Only at boiling point
- (k) Throughout the liquid only

- (j) At the surface of a liquid
- (l) Only below 0 degrees C

Answer: _____

3. Boiling differs from evaporation because boiling:

- (i) Occurs only at the surface
- (k) Can occur at any temperature

- (j) Occurs throughout the liquid at a fixed temperature
- (l) Needs no heat

Answer: _____

4. Heat is transferred mainly through a metal rod by:

- (i) Conduction
- (k) Radiation

- (j) Convection
- (l) Evaporation

Answer: _____

5. Heat transfer in liquids and gases mainly occurs by:

- (i) Conduction only
- (k) Reflection

- (j) Convection
- (l) Dispersion

Answer: _____

6. The Sun warms the Earth mainly by:

- (i) Conduction
- (k) Radiation

- (j) Convection
- (l) Evaporation

Answer: _____

7. A dull black surface is generally a:

- (i) Poor absorber and poor emitter
- (k) Good reflector only

- (j) Good absorber and good emitter
- (l) Perfect insulator

Answer: _____

8. Small gaps are left between railway tracks mainly to allow for:

- (i) Linear expansion
- (k) Cooling by convection

- (j) Superficial expansion only
- (l) Radiation loss

Answer: _____

SECTION B - 12 MARKS

Answer all questions. Continue on the back side if additional space is required.

9. Differentiate between evaporation and boiling using any two points. [2]

10. Explain how heat travels from the hot end to the cold end of a metal spoon. [2]

11. Explain briefly how convection currents are set up when water is heated from below. [2]

12. Explain the formation of a sea breeze during the daytime. [2]

13. What are linear, superficial and volume expansion? Give one example of any one of them. [2]

14. Why are cooking utensils made of metal but their handles are often made of wood or plastic? [2]

ASCEND INSTITUTE, GANGTOK

CHAPTER-WISE TEST

ICSE CLASS 8 - BIOLOGY

Chapter: Human Body - I: Control and Coordination

Date: _____

Name: _____

Maximum Marks: 20

Time: 40 minutes

Instructions: Answer all questions. Section A contains 8 MCQs of 1 mark each. Section B contains 6 questions of 2 marks each. Show the necessary steps wherever required.

SECTION A - 8 MARKS

1. The neuron that carries impulses from a sense organ to the brain or spinal cord is a:

- | | |
|------------------------|--------------------|
| (i) Motor neuron | (j) Sensory neuron |
| (k) Association neuron | (l) Mixed nerve |

Answer: _____

2. The part of the brain that mainly maintains balance and coordinates muscular activity is the:

- | | |
|-----------------------|-----------------|
| (i) Cerebrum | (j) Cerebellum |
| (k) Medulla oblongata | (l) Spinal cord |

Answer: _____

3. The medulla oblongata mainly controls:

- | | |
|---|-----------------------|
| (i) Thinking only | (j) Voluntary writing |
| (k) Involuntary actions such as breathing | (l) Growth of bones |

Answer: _____

4. The pituitary is called the master gland because it:

- | | |
|------------------------------|-------------------------------------|
| (i) Produces digestive juice | (j) Controls other endocrine glands |
| (k) Carries nerve impulses | (l) Forms red blood cells |

Answer: _____

5. Which gland has both endocrine and exocrine functions?

- | | |
|--------------|---------------|
| (i) Thyroid | (j) Pituitary |
| (k) Pancreas | (l) Adrenal |

Answer: _____

6. The hormone that prepares the body for an emergency is:

- | | |
|---------------|----------------|
| (i) Insulin | (j) Adrenaline |
| (k) Thyroxine | (l) Glucagon |

Answer: _____

7. Many quick reflex actions are controlled mainly by the:

- | | |
|----------------|-----------------|
| (i) Cerebrum | (j) Spinal cord |
| (k) Cerebellum | (l) Pituitary |

Answer: _____

8. Which is a common change during adolescence?

- | | |
|-----------------------------------|----------------------------|
| (i) Reproductive organs mature | (j) All growth stops |
| (k) Heartbeat permanently doubles | (l) Sense organs disappear |

Answer: _____

SECTION B - 12 MARKS

Answer all questions. Continue on the back side if additional space is required.

9. State two differences between nervous control and hormonal control. [2]

10. Write the correct sequence of a reflex pathway from receptor to effector. [2]

11. State one function each of the cerebrum, cerebellum and medulla oblongata. [2]

12. Write any two functions of the spinal cord. [2]

13. Name one hormone secreted by each of these glands and state one function: pituitary and pancreas. [2]

14. State any four physical changes commonly seen during adolescence. [2]

ASCEND INSTITUTE, GANGTOK

CHAPTER-WISE TEST

ICSE CLASS 8 - CHEMISTRY

Chapter: Language of Chemistry

Date: _____

Name: _____

Maximum Marks: 20

Time: 40 minutes

Instructions: Answer all questions. Section A contains 8 MCQs of 1 mark each. Section B contains 6 questions of 2 marks each. Show the necessary steps wherever required.

SECTION A - 8 MARKS

1. The correct formula formed by Na^+ and O^{2-} is:

(i) NaO

(k) NaO_2

(j) Na_2O

(l) Na_2O_2

Answer: _____

2. The correct formula of aluminium sulphate is:

(i) AlSO_4

(k) $\text{Al}_2(\text{SO}_4)_3$

(j) Al_2SO_4

(l) $\text{Al}_3(\text{SO}_4)_2$

Answer: _____

3. Calcium has valency 2 and hydroxide has valency 1. The formula of calcium hydroxide is:

(i) CaOH

(k) $\text{Ca}(\text{OH})_2$

(j) Ca_2OH

(l) $\text{Ca}_2(\text{OH})_2$

Answer: _____

4. How many oxygen atoms are present in $\text{Ca}(\text{NO}_3)_2$?

(i) 2

(k) 5

(j) 3

(l) 6

Answer: _____

5. The formula Al_2O_3 is consistent with valencies:

(i) $\text{Al} = 2$, $\text{O} = 3$

(k) $\text{Al} = 1$, $\text{O} = 3$

(j) $\text{Al} = 3$, $\text{O} = 2$

(l) $\text{Al} = 3$, $\text{O} = 1$

Answer: _____

6. While balancing an equation, one should change:

(i) Only subscripts

(k) The formula of each compound

(j) Only coefficients

(l) Names of substances

Answer: _____

7. Which equation is balanced?

(i) $\text{H}_2 + \text{O}_2 \rightarrow \text{H}_2\text{O}$

(k) $\text{H}_2 + 2\text{O}_2 \rightarrow \text{H}_2\text{O}$

(j) $2\text{H}_2 + \text{O}_2 \rightarrow 2\text{H}_2\text{O}$

(l) $2\text{H}_2 + 2\text{O}_2 \rightarrow \text{H}_2\text{O}$

Answer: _____

8. The correct formula of ammonium sulphate is:

(i) NH_4SO_4

(k) $\text{NH}_4(\text{SO}_4)_2$

(j) $(\text{NH}_4)_2\text{SO}_4$

(l) $(\text{NH}_4)_2(\text{SO}_4)_2$

Answer: _____

SECTION B - 12 MARKS

Answer all questions. Continue on the back side if additional space is required.

9. Write the formula of aluminium sulphate using the valencies $\text{Al} = 3$ and $\text{SO}_4 = 2$. [2]

10. Write the formula of magnesium hydroxide using $\text{Mg} = 2$ and $\text{OH} = 1$. [2]

11. In Fe_2O_3 , oxygen has valency 2. Find the valency of iron. [2]

12. Find the total number of atoms of each element in $\text{Al}_2(\text{SO}_4)_3$. [2]

13. Balance: $\text{H}_2 + \text{O}_2 \rightarrow \text{H}_2\text{O}$. [2]

14. A student writes magnesium chloride as MgCl . Correct the formula and explain why. [2]

ASCEND INSTITUTE, GANGTOK

CHAPTER-WISE TEST

ICSE CLASS 8 - CHEMISTRY

Chapter: Chemical Reactions and Equations

Date: _____

Name: _____

Maximum Marks: 20

Time: 40 minutes

Instructions: Answer all questions. Section A contains 8 MCQs of 1 mark each. Section B contains 6 questions of 2 marks each. Show the necessary steps wherever required.

SECTION A - 8 MARKS

1. When iron reacts with oxygen to form iron oxide, it is mainly a:

- | | |
|---------------------------|----------------------------------|
| (i) Combination reaction | (j) Decomposition reaction |
| (k) Displacement reaction | (l) Double displacement reaction |

Answer: _____

2. When copper sulphate reacts with sodium hydroxide and a blue insoluble solid forms, the reaction is:

- | | |
|------------------------|---|
| (i) Only displacement | (j) Precipitation and double displacement |
| (k) Only decomposition | (l) Photochemical |

Answer: _____

3. A metal above another metal in the reactivity series can generally:

- | | |
|--|-------------------------------------|
| (i) Displace the lower metal from its salt | (j) Be displaced by the lower metal |
| (k) Never react | (l) Only form gases |

Answer: _____

4. Which oxide is amphoteric?

- | | |
|-----------------------------|-------------------|
| (i) SO_2 | (j) SO_3 |
| (k) Al_2O_3 | (l) CO_2 |

Answer: _____

5. Heating a metal carbonate strongly in little or no air to form its oxide is called:

- | | |
|--------------------|-------------------|
| (i) Roasting | (j) Calcination |
| (k) Neutralization | (l) Precipitation |

Answer: _____

6. Heating a metal sulphide strongly in excess air is called:

- | | |
|--------------------|-----------------|
| (i) Calcination | (j) Roasting |
| (k) Neutralization | (l) Combination |

Answer: _____

7. A reaction that absorbs heat from the surroundings is:

- | | |
|----------------|--------------------|
| (i) Exothermic | (j) Endothermic |
| (k) Neutral | (l) Catalytic only |

Answer: _____

8. A substance that changes the rate of a reaction without being used up is a:

- | | |
|--------------|-----------------|
| (i) Reactant | (j) Product |
| (k) Catalyst | (l) Precipitate |

Answer: _____

SECTION B - 12 MARKS

Answer all questions. Continue on the back side if additional space is required.

9. Classify and balance: $\text{Fe} + \text{CuSO}_4 \rightarrow \text{FeSO}_4 + \text{Cu}$. [2]

10. Predict whether Cu will displace Zn from ZnSO_4 . Give a reason using the reactivity series. [2]

11. Complete and balance: $\text{AgNO}_3 + \text{NaCl} \rightarrow \text{_____} + \text{_____}$. Name the precipitate. [2]

12. What is a neutralization reaction? Write one balanced example. [2]

13. Differentiate between calcination and roasting. [2]

14. State any two signs that indicate that a chemical reaction has taken place. [2]

ASCEND INSTITUTE, GANGTOK
CHAPTER-WISE TEST
ICSE CLASS 10 - CHEMISTRY

Chapter: Organic Chemistry

Date: _____

Name: _____

Maximum Marks: 20

Time: 45 minutes

Instructions: Answer all questions. Section A contains 8 MCQs of 1 mark each. Section B contains 6 questions of 2 marks each. Show the necessary steps wherever required.

SECTION A - 8 MARKS

1. Which pair represents chain isomers?

- (i) Pentane and 2-methylbutane
(k) Ethane and propane

- (j) But-1-yne and but-2-yne
(l) Propene and propyne

Answer: _____

2. An open-chain hydrocarbon with formula C_6H_{10} belongs to the series:

- (i) C_nH_{2n+2}
(k) C_nH_{2n-2}

- (j) C_nH_{2n}
(l) C_nH_{2n+1}

Answer: _____

3. The IUPAC name of $CH_3-CH(CH_3)-CH(Br)-CH_2-CH_3$ is:

- (i) 3-bromo-2-methylpentane
(k) 2-methyl-3-bromopentane

- (j) 3-bromo-4-methylpentane
(l) 3-bromo-2-methylhexane

Answer: _____

4. Which pair can be successive members of the same homologous series?

- (i) C_3H_8 and C_4H_{10}
(k) C_4H_8 and C_5H_{12}

- (j) C_3H_6 and C_4H_{10}
(l) C_2H_2 and C_3H_6

Answer: _____

5. Which functional group is present in alcohols?

- (i) -CHO
(k) -OH

- (j) -COOH
(l) -CO-

Answer: _____

6. $CH_3-CH=CH-CH_3$ is named:

- (i) But-1-ene
(k) Butane

- (j) But-2-ene
(l) But-2-yne

Answer: _____

7. Compounds with the same molecular formula but different structural formulae are called:

- (i) Homologues
(k) Radicals

- (j) Isomers
(l) Catalysts

Answer: _____

8. Which hydrocarbon generally undergoes addition reactions?

- (i) Alkane
(k) Saturated hydrocarbon only

- (j) Alkene
(l) Methane only

Answer: _____

SECTION B - 12 MARKS

Answer all questions. Continue on the back side if additional space is required.

9. Give the IUPAC name of $\text{CH}_3\text{-C}(\text{Cl})(\text{CH}_3)\text{-CH}_2\text{-CH}(\text{Br})\text{-CH}_3$. [2]

10. Give the IUPAC name of $\text{CH}_3\text{-CH=C}(\text{CH}_3)\text{-CH}(\text{Cl})\text{-CH}_3$. [2]

11. Write the structural formula of 3-bromo-2-methylpentane. [2]

12. Draw or write the condensed structural formulae of all three chain isomers of C_5H_{12} . [2]

13. State the homologous series and general formula for C_4H_{10} , C_4H_8 and C_4H_6 . [2]

14. Why do alkanes generally undergo substitution reactions, while alkenes and alkynes undergo addition reactions? [2]

ASCEND INSTITUTE, GANGTOK
CHAPTER-WISE TEST
CBSE CLASS 6 - MATHEMATICS

Chapter: Prime Numbers

Date: _____

Name: _____

Maximum Marks: 20

Time: 40 minutes

Instructions: Answer all questions. Section A contains 8 MCQs of 1 mark each. Section B contains 6 questions of 2 marks each. Show the necessary steps wherever required.

SECTION A - 8 MARKS

1. Which of the following is a prime number?

- | | |
|--------|--------|
| (i) 51 | (j) 57 |
| (k) 59 | (l) 69 |

Answer: _____

2. Which number is composite?

- | | |
|--------|--------|
| (i) 29 | (j) 31 |
| (k) 37 | (l) 39 |

Answer: _____

3. The prime factorisation of 84 is:

- | | |
|------------------------------------|----------------------------|
| (i) $2 \times 2 \times 3 \times 7$ | (j) $2 \times 3 \times 14$ |
| (k) 4×21 | (l) 2×42 |

Answer: _____

4. Which pair is co-prime?

- | | |
|---------------|---------------|
| (i) 8 and 15 | (j) 6 and 9 |
| (k) 10 and 15 | (l) 12 and 18 |

Answer: _____

5. Which is the smallest prime number?

- | | |
|-------|-------|
| (i) 0 | (j) 1 |
| (k) 2 | (l) 3 |

Answer: _____

6. How many prime numbers are there between 10 and 20?

- | | |
|-------|-------|
| (i) 2 | (j) 3 |
| (k) 4 | (l) 5 |

Answer: _____

7. Which number has exactly three prime factors when repeated factors are counted?

- | | |
|--------|--------|
| (i) 12 | (j) 13 |
| (k) 17 | (l) 19 |

Answer: _____

8. Which statement is true?

- | | |
|-------------------------------|-------------------------------------|
| (i) 1 is a prime number | (j) 2 is the only even prime number |
| (k) Every odd number is prime | (l) Every prime number is odd |

Answer: _____

SECTION B - 12 MARKS

Answer all questions. Continue on the back side if additional space is required.

9. Express 90 as a product of prime factors. [2]

10. Is 91 prime or composite? Write a factor pair to support your answer. [2]

11. List all prime numbers between 30 and 50. [2]

12. Express 180 as a product of prime factors. [2]

13. Write any two pairs of twin primes less than 30. [2]

14. Find the common prime factors of 60 and 84. [2]

ASCEND INSTITUTE, GANGTOK
CHAPTER-WISE TEST
CBSE CLASS 6 - MATHEMATICS

Chapter: Fractions

Date: _____

Name: _____

Maximum Marks: 20

Time: 40 minutes

Instructions: Answer all questions. Section A contains 8 MCQs of 1 mark each. Section B contains 6 questions of 2 marks each. Show the necessary steps wherever required.

SECTION A - 8 MARKS

1. Which fraction is equivalent to $\frac{3}{5}$?

- (i) $\frac{6}{15}$
(k) $\frac{12}{25}$

- (j) $\frac{9}{15}$
(l) $\frac{15}{20}$

Answer: _____

2. Which is greater?

- (i) $\frac{7}{12}$
(k) Both equal

- (j) $\frac{5}{8}$
(l) Cannot be compared

Answer: _____

3. Which is a proper fraction?

- (i) $\frac{7}{5}$
(k) $\frac{5}{7}$

- (j) $\frac{9}{8}$
(l) $\frac{11}{6}$

Answer: _____

4. $\frac{2}{3} + \frac{1}{6} =$

- (i) $\frac{1}{2}$
(k) $\frac{5}{6}$

- (j) $\frac{2}{9}$
(l) 1

Answer: _____

5. $\frac{7}{8} - \frac{3}{8} =$

- (i) $\frac{1}{2}$
(k) $\frac{4}{5}$

- (j) $\frac{3}{8}$
(l) $\frac{5}{8}$

Answer: _____

6. The mixed number $2\frac{3}{5}$ equals:

- (i) $\frac{10}{5}$
(k) $\frac{13}{5}$

- (j) $\frac{11}{5}$
(l) $\frac{15}{5}$

Answer: _____

7. $\frac{3}{4}$ of 20 is:

- (i) 5
(k) 15

- (j) 10
(l) 16

Answer: _____

8. Which fraction is in simplest form?

- (i) $\frac{6}{8}$
(k) $\frac{5}{7}$

- (j) $\frac{9}{12}$
(l) $\frac{12}{18}$

Answer: _____

SECTION B - 12 MARKS

Answer all questions. Continue on the back side if additional space is required.

9. Simplify $24/36$ to its lowest form. [2]

10. Add: $5/8 + 1/4$. [2]

11. Subtract: $7/9 - 2/3$. [2]

12. Convert $2\frac{3}{5}$ into an improper fraction. [2]

13. Rina read $3/8$ of a book in the morning and $1/4$ in the evening. What fraction did she read altogether? [2]

14. Compare $7/12$ and $5/8$ using a suitable method. [2]

ASCEND INSTITUTE, GANGTOK
CHAPTER-WISE TEST
CBSE CLASS 7 - MATHEMATICS

Chapter: Integers

Date: _____

Name: _____

Maximum Marks: 20

Time: 40 minutes

Instructions: Answer all questions. Section A contains 8 MCQs of 1 mark each. Section B contains 6 questions of 2 marks each. Show the necessary steps wherever required.

SECTION A - 8 MARKS

1. $-8 + 15 =$

(i) -23

(j) -7

(k) 7

(l) 23

Answer: _____

2. $(-6) \times (-4) =$

(i) -24

(j) -10

(k) 10

(l) 24

Answer: _____

3. The additive inverse of -12 is:

(i) -12

(j) 0

(k) 12

(l) $1/12$

Answer: _____

4. $(-35) \div 7 =$

(i) -5

(j) 5

(k) -28

(l) 28

Answer: _____

5. Which is greatest?

(i) -9

(j) -4

(k) -12

(l) -7

Answer: _____

6. $-3 - (-8) =$

(i) -11

(j) -5

(k) 5

(l) 11

Answer: _____

7. The product of three negative integers is always:

(i) Positive

(j) Negative

(k) Zero

(l) Cannot be predicted

Answer: _____

8. At 6 a.m. the temperature is -4 degrees C and by noon it is 7 degrees C. The rise is:

(i) 3 degrees C

(j) 7 degrees C

(k) 11 degrees C

(l) -11 degrees C

Answer: _____

SECTION B - 12 MARKS

Answer all questions. Continue on the back side if additional space is required.

9. Simplify: $-18 + 7 - (-5)$. [2]

10. Evaluate: $(-24) \div 6 + (-3) \times 4$. [2]

11. Find: $(-15) + 28 - 19$. [2]

12. Evaluate: $(-8) \times 7 + 20$. [2]

13. A submarine is 35 m below sea level and rises 18 m. Represent the final position using an integer. [2]

14. Use the distributive property to evaluate: $(-7) \times (12 - 5)$. [2]

ASCEND INSTITUTE, GANGTOK
CHAPTER-WISE TEST
CBSE CLASS 7 - MATHEMATICS

Chapter: Rational Numbers

Date: _____

Name: _____

Maximum Marks: 20

Time: 40 minutes

Instructions: Answer all questions. Section A contains 8 MCQs of 1 mark each. Section B contains 6 questions of 2 marks each. Show the necessary steps wherever required.

SECTION A - 8 MARKS

1. Which rational number is greatest?

(i) $-\frac{3}{5}$

(j) $-\frac{2}{3}$

(k) $-\frac{7}{10}$

(l) $-\frac{4}{5}$

Answer: _____

2. The additive inverse of $-\frac{7}{9}$ is:

(i) $-\frac{9}{7}$

(j) $\frac{7}{9}$

(k) $\frac{9}{7}$

(l) 0

Answer: _____

3. The reciprocal of $-\frac{5}{8}$ is:

(i) $\frac{5}{8}$

(j) $-\frac{8}{5}$

(k) $\frac{8}{5}$

(l) $-\frac{5}{8}$

Answer: _____

4. $-\frac{3}{4} + \frac{5}{8} =$

(i) $-\frac{1}{8}$

(j) $\frac{1}{8}$

(k) $-\frac{11}{8}$

(l) $\frac{11}{8}$

Answer: _____

5. Which is in standard form?

(i) $\frac{6}{8}$

(j) $-\frac{9}{12}$

(k) $\frac{5}{7}$

(l) $\frac{10}{15}$

Answer: _____

6. $\frac{7}{12} - (-\frac{5}{18}) =$

(i) $\frac{11}{36}$

(j) $\frac{31}{36}$

(k) $\frac{1}{36}$

(l) $-\frac{31}{36}$

Answer: _____

7. Which rational number lies between $\frac{1}{2}$ and $\frac{3}{4}$?

(i) $\frac{1}{4}$

(j) $\frac{5}{8}$

(k) $\frac{4}{5}$

(l) $\frac{7}{8}$

Answer: _____

8. The product of a non-zero rational number and its reciprocal is:

(i) 0

(j) 1

(k) -1

(l) The same number

Answer: _____

SECTION B - 12 MARKS

Answer all questions. Continue on the back side if additional space is required.

9. Reduce $-18/24$ to standard form. [2]

10. Arrange in ascending order: $-2/3$, $1/2$, $-5/6$, 0 . [2]

11. Simplify: $7/12 - (-5/18)$. [2]

12. Find: $(-3/5) \times (10/9)$. [2]

13. Divide: $(-7/8) \div (14/15)$. [2]

14. Find a rational number between $2/5$ and $3/5$ and show how you obtained it. [2]

ASCEND INSTITUTE, GANGTOK
CHAPTER-WISE TEST
CBSE CLASS 8 - MATHEMATICS

Chapter: Perimeters and Areas

Date: _____

Name: _____

Maximum Marks: 20

Time: 40 minutes

Instructions: Answer all questions. Section A contains 8 MCQs of 1 mark each. Section B contains 6 questions of 2 marks each. Show the necessary steps wherever required.

SECTION A - 8 MARKS

1. The perimeter of a rectangle of length 12 cm and breadth 7 cm is:

- | | |
|-----------|-----------|
| (i) 19 cm | (j) 38 cm |
| (k) 84 cm | (l) 24 cm |

Answer: _____

2. The area of a parallelogram with base 9 cm and height 6 cm is:

- | | |
|-----------------------|-----------------------|
| (i) 15 cm^2 | (j) 27 cm^2 |
| (k) 54 cm^2 | (l) 90 cm^2 |

Answer: _____

3. A square has area 144 cm^2 . Its perimeter is:

- | | |
|-----------|------------|
| (i) 24 cm | (j) 36 cm |
| (k) 48 cm | (l) 144 cm |

Answer: _____

4. The area of a triangle with base 10 cm and height 8 cm is:

- | | |
|-----------------------|-----------------------|
| (i) 18 cm^2 | (j) 40 cm^2 |
| (k) 80 cm^2 | (l) 90 cm^2 |

Answer: _____

5. If the length of a rectangle doubles while breadth stays the same, its area becomes:

- | | |
|------------|----------------|
| (i) Half | (j) Same |
| (k) Double | (l) Four times |

Answer: _____

6. A square has perimeter 36 cm. Its side is:

- | | |
|-----------|-----------|
| (i) 6 cm | (j) 9 cm |
| (k) 12 cm | (l) 18 cm |

Answer: _____

7. A rectangle has area 96 cm^2 and length 12 cm. Its breadth is:

- | | |
|-----------|-----------|
| (i) 6 cm | (j) 8 cm |
| (k) 10 cm | (l) 12 cm |

Answer: _____

8. Which unit is suitable for area?

- | | |
|-------------------|-------------------|
| (i) cm | (j) cm^2 |
| (k) cm^3 | (l) kg |

Answer: _____

SECTION B - 12 MARKS

Answer all questions. Continue on the back side if additional space is required.

9. Find the perimeter and area of a rectangle of length 15 cm and breadth 8 cm. [2]

10. A square has side 11 cm. Find its perimeter and area. [2]

11. Find the area of a parallelogram of base 14 cm and height 9 cm. [2]

12. Find the area of a triangle of base 18 cm and height 7 cm. [2]

13. A rectangular garden is 20 m long and 14 m wide. Find the length of fencing needed around it. [2]

14. The area of a rectangle is 180 cm^2 and its breadth is 12 cm. Find its length and perimeter. [2]

ASCEND INSTITUTE, GANGTOK
CHAPTER-WISE TEST
CBSE CLASS 8 - MATHEMATICS

Chapter: Rational Numbers

Date: _____

Name: _____

Maximum Marks: 20

Time: 40 minutes

Instructions: Answer all questions. Section A contains 8 MCQs of 1 mark each. Section B contains 6 questions of 2 marks each. Show the necessary steps wherever required.

SECTION A - 8 MARKS

1. The reciprocal of $-7/11$ is:

(i) $7/11$

(j) $-11/7$

(k) $11/7$

(l) $-7/11$

Answer: _____

2. $-3/4 \div 9/10 =$

(i) $-5/6$

(j) $5/6$

(k) $-27/40$

(l) $27/40$

Answer: _____

3. The additive inverse of $5/9$ is:

(i) $-5/9$

(j) $9/5$

(k) $5/9$

(l) 0

Answer: _____

4. Which is greater?

(i) $-2/3$

(j) $-3/5$

(k) Both equal

(l) Cannot be compared

Answer: _____

5. The product of $-4/7$ and $21/8$ is:

(i) $-3/2$

(j) $3/2$

(k) $-12/56$

(l) $12/56$

Answer: _____

6. Which property is shown by $a + b = b + a$?

(i) Closure

(j) Commutative

(k) Associative

(l) Distributive

Answer: _____

7. Which rational number lies between $1/3$ and $1/2$?

(i) $1/4$

(j) $5/12$

(k) $2/3$

(l) $3/4$

Answer: _____

8. For any rational number a , $a \times 1 =$

(i) 0

(j) 1

(k) a

(l) $-a$

Answer: _____

SECTION B - 12 MARKS

Answer all questions. Continue on the back side if additional space is required.

9. Simplify: $-\frac{5}{6} + \frac{7}{9}$. [2]

10. Evaluate: $(-\frac{3}{4}) \div (\frac{9}{10})$. [2]

11. Arrange in ascending order: $-\frac{3}{4}$, $\frac{2}{5}$, $-\frac{1}{2}$, 0. [2]

12. Use the distributive property to evaluate: $\frac{3}{5} \times (\frac{10}{9} - \frac{5}{6})$. [2]

13. Find the reciprocal and additive inverse of $-\frac{7}{12}$. [2]

14. Find three rational numbers between $\frac{2}{5}$ and $\frac{3}{5}$. [2]
